

MEGAN L. KRAUSE
MECHANICAL DESIGN ENGINEER

PROFESSIONAL SUMMARY

Results-driven mechanical design engineer with 3 years of experience in product development and manufacturing. Skilled in creating detailed 3D models, conducting tolerance stack-up analysis, and ensuring designs meet functional and production requirements. Adept at collaborating with cross-functional teams to deliver high-quality, cost-effective solutions.

EDUCATION

BACHELOR OF SCIENCE IN MECHANICAL ENGINEERING
UNIVERSITY OF MINNESOTA - TWIN CITIES
GRADUATED: JUNE 2020

SKILLS

SOLIDWORKS, AUTOCAD, TOLERANCE STACK-UP ANALYSIS, CAD MODELING, PRODUCT DESIGN, MANUFACTURING PROCESS PLANNING, DIMENSIONAL TOLERANCE SPECIFICATION, GEOMETRIC TOLERANCING, DRAFTING, 3D PRINTING, PRODUCT ASSEMBLY, INTERPRETING BLUEPRINTS, DESIGN FOR MANUFACTURING, MECHANICAL DRAWING, CAD SOFTWARE, MECHATRONICS, DESIGN REVIEW, QUALITY CONTROL

EXPERIENCE

MECHANICAL DESIGN ENGINEER
TECHNOLOGY COMPONENTS INC.
MINNEAPOLIS, MN | SEP 2020 - PRESENT

- Designed and optimized mechanical components for consumer electronics, reducing assembly time by 15% through improved part integration.
- Conducted tolerance stack-up analysis on critical assemblies to ensure dimensional accuracy and prevent assembly failures.
- Collaborated with manufacturing teams to refine design for production, resulting in a 20% reduction in material waste.
- Created detailed 2D and 3D drawings using AutoCAD and SolidWorks, ensuring compliance with industry standards and client specifications.

MECHANICAL DESIGN ENGINEER
INNOVATE FABRICATIONS LLC
ST. PAUL, MN | JAN 2020 - AUG 2020

- Led the design of custom mechanical systems for industrial applications, focusing on precision and manufacturability.
- Performed tolerance analysis on critical mating surfaces to ensure proper fit and function under operational stress.
- Prepared design documentation and technical drawings for internal review and client submission, enhancing communication across departments.

- Supported product development lifecycle from concept to prototype, ensuring design feasibility and alignment with production capabilities.

PROJECTS

AUTOMATED PACKAGING SYSTEM DESIGN

DEVELOPED A MECHANICAL SYSTEM FOR AN AUTOMATED PACKAGING LINE, FOCUSED ON IMPROVING EFFICIENCY AND REDUCING MAINTENANCE REQUIREMENTS. THE DESIGN INCORPORATED MODULAR COMPONENTS AND STANDARDIZED GEOMETRIC TOLERANCES TO ENSURE EASY ASSEMBLY AND LONG-TIME RELIABILITY.

TOLERANCE ANALYSIS FOR PRECISION MECHANICAL ASSEMBLY

PERFORMED TOLERANCE STACK-UP ANALYSIS ON A HIGH-PRECISION MECHANICAL ASSEMBLY TO IDENTIFY POTENTIAL DIMENSIONAL ISSUES AND OPTIMIZE TOLERANCE VALUES. THE ANALYSIS RESULTED IN A 12% IMPROVEMENT IN ASSEMBLY SUCCESS RATE AND REDUCED COSTS ASSOCIATED WITH REWORK.

CERTIFICATIONS

NONE